

Linear Models Lecture 7: Classic Normal Regression (small sample)

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Where Are We? *Efficient Estimation*

- So far: OLS estimates β and we have its variance, but *how precise can any estimator be?*
- The general theory of **efficient estimation** answers this via **scores** (derivatives of log-densities) and **information** (their variance).
- These concepts are not specific to MLE — they underlie influence functions, semiparametric efficiency bounds, and GMM.
- **This lecture:** when we know the full likelihood, MLE is the estimator that achieves the efficiency bound. We use the normal linear model as the leading case.
- **The punchline:** OLS *is* the MLE under normality, and its score equations are moment conditions — the same objects that drive GMM and robust inference.

Parametric Model

Definition

A **parametric model** is a family of probability distributions $\{f(x|\theta) : \theta \in \Theta\}$ indexed by a finite-dimensional parameter vector θ .

- In the continuous case, $f(x|\theta)$ is a probability density function.
- E.g. if $X \sim N(\mu, \sigma^2)$, $f(x|\mu, \sigma^2) = \sigma^{-1}\phi((x - \mu)/\sigma)$, the parameters are $\mu \in \mathbb{R}$ and $\sigma^2 > 0$.
- Recall, $\phi(z)$ is the density for the standard normal, $\phi(z) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{z^2}{2})$

Specified Parametric Models

- A model is called **correctly specified** when there exists a unique parameter value θ_0 such that $f(x|\theta_0) = f(x)$, the true data distribution.
- For example, if the true density is

$$f(x) = 2 \exp(-2x)$$

- the exponential model $f(x|\lambda) = \lambda^{-1} \exp(-x/\lambda)$ is a correctly specified model with $\lambda_0 = 1/2$
- the lognormal model $f(x|\mu, \sigma^2) = \frac{1}{x\sigma\sqrt{2\pi}} \exp(-\frac{(\ln x - \mu)^2}{2\sigma^2})$ is misspecified, cannot equal $2 \exp(-2x)$ under any parameter value.

Likelihoods

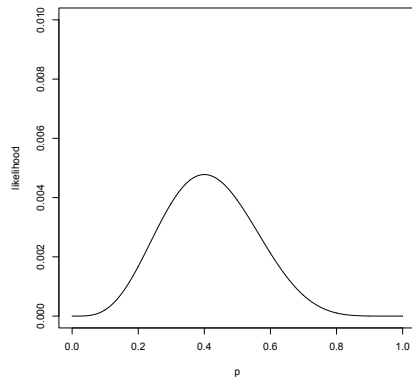
- The density $f(x|\theta)$ tells us: for a fixed θ , how probable is a particular observation x ?
- The **likelihood** asks the reverse question: for fixed observed data $\mathbf{X}$, how plausible is each θ ?

$$\mathcal{L}(\theta) \equiv f(X_1, \dots, X_n | \theta) \quad (\text{same formula, viewed as a function of } \theta)$$

- Example: Binomial model of how many terms (x out of $n = 2$) each of 5 candidates wins.
 $P(x|p) = \binom{n}{x} p^x (1-p)^{n-x}$.
- Suppose we observe $\mathbf{x} = \{1, 0, 1, 2, 0\}$.
- The Joint Likelihood of the data is:

$$\begin{aligned} P(\mathbf{x} | p) &= \binom{2}{1} p^1 (1-p)^1 \binom{2}{0} p^0 (1-p)^2 \binom{2}{1} p^1 (1-p)^1 \binom{2}{2} p^2 (1-p)^0 \binom{2}{0} p^0 (1-p)^2 \\ &= 4p^4 (1-p)^6 \end{aligned}$$

Likelihood for Binomial x



Maximum Likelihood Estimator (from Hansen Probability)

Definition

The **maximum likelihood estimator** $\hat{\theta}$ of θ is the value that maximizes the likelihood:

$$\mathcal{L}_n(\theta) \equiv f(X_1, X_2, \dots, X_n | \theta)$$

Call $\ell_n = \sum_{i=1}^n \log f(X_i | \theta)$ the log-likelihood.

Maximizing the likelihood

$$P(\mathbf{x} | p) = 4p^4(1 - p)^6$$

$$\frac{\partial}{\partial p} P(\mathbf{x} | p) = 16p^3(1 - p)^6 - 24p^4(1 - p)^5$$

$$= 0$$

$$3p^4(1 - p)^5 = 2p^3(1 - p)^6$$

$$3p = 2(1 - p)$$

$$5p = 2$$

$$p^* = \frac{2}{5}$$

From MLE to Precision

We now have an estimator (MLE) and a useful property (invariance). **Two questions remain:**

- 1 How precise is the MLE?** We need a measure of information in the data.
- 2 How precise can any estimator be?** We need a lower bound on variance.

The answers come from three linked concepts:

- **Score:** the derivative of the log-density (the “gradient” that MLE sets to zero).
- **Fisher Information:** the variance of the score (how much the data tells us).
- **Cramér-Rao Bound:** no unbiased estimator can beat information⁻¹.

Score: the slope of the log likelihood

- The **likelihood score** is the derivative of the log-likelihood function:

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta)$$

- The score is a function of θ and tells us how sensitive the log-likelihood is to the parameter, and equals zero at the optimum.
- The **efficient score** is the derivative of the log likelihood for a single observation, evaluated at $\mathbf{x} = X_1, X_2, \dots, X_n$ and the true parameter vector

$$S = \frac{\partial}{\partial \theta} \log f(\mathbf{x}|\theta_0)$$

- The efficient score fixes θ at the true value θ_0 .

Hessian: the curvature of the log likelihood

- The **likelihood Hessian** is the negative second derivative:

$$\mathcal{H}_n(\theta) = -\frac{\partial^2}{\partial\theta\partial\theta'}\ell_n(\theta)$$

- The Hessian matrix is used to calculate the variance.

Fisher Information

- The **Fisher information** is the variance of the efficient score (score evaluated at true θ_0).

$$\mathcal{I}_\theta = \mathbb{E}[SS']$$

- The **expected Hessian** is the expectation of the Hessian for a single observation:

$$\mathcal{H}_\theta = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta\partial\theta'} \log f(X|\theta)\right]$$

- If the model is correctly specified:

$$\mathcal{I}_\theta = \mathcal{H}_\theta$$

- **Intuition:** At θ_0 , the score has mean zero. Both $\mathcal{I}$ (score variance) and $\mathcal{H}$ (expected curvature) measure second-order properties of the same density—they agree because these are two ways of measuring the same thing.

Cramér-Rao Lower Bound

- If $\tilde{\theta}$ is *any* unbiased estimator of θ , then:

$$\text{Var}[\tilde{\theta}] \geq (n\mathcal{I}_{\theta})^{-1}$$

No unbiased estimator can have variance below the inverse of the total Fisher information.

- **Intuition:** More informative data (higher $\mathcal{I}_{\theta}$) $\rightarrow$ tighter bound $\rightarrow$ more precise estimation is possible. The bound is an information-theoretic limit on what the data can tell you.
- An estimator that achieves the bound is called **efficient**.

Cramér-Rao vs. Gauss-Markov

- **Gauss-Markov** (Lecture 5): OLS is the best *linear* unbiased estimator, under homoskedasticity. A result specific to the linear model.
- **Cramér-Rao**: No unbiased estimator—linear or not—can beat the information bound. A general result for *any parametric model*.
- Under normality, OLS achieves the CR bound → efficient among **all** unbiased estimators, not just linear ones.
- **The broader point**: The CR bound is one instance of a *semiparametric efficiency bound*. In later lectures, we will see that the same logic—scores, information, influence functions—defines efficient estimators even without a full parametric model.

Variance Estimators

- The sample Hessian Estimator depends on calculating the second derivatives of the log-likelihood:

$$\hat{\mathcal{H}}_{\theta} = \frac{1}{n} \sum_{i=1}^n -\frac{\partial}{\partial \theta \partial \theta'} \log f(X_i | \hat{\theta})$$
$$\hat{\mathbf{V}}_1 = \hat{\mathcal{H}}_{\theta}^{-1}$$

- The Outer Product Estimator is based on the Fisher Information:

$$\hat{\mathcal{G}}_{\theta} = \frac{1}{n} \sum_{i=1}^n \left(\frac{\partial}{\partial \theta} \log f(X_i | \hat{\theta}) \right) \left(\frac{\partial}{\partial \theta} \log f(X_i | \hat{\theta}) \right)'$$
$$\hat{\mathbf{V}}_2 = \hat{\mathcal{G}}_{\theta}^{-1}$$

Example: Normal with known mean, unknown variance

$X \sim N(0, \theta)$, where $\theta \equiv \sigma^2$. $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = \theta_0$, $\mathbb{E}[X^4] = 3\theta_0^2$.

The density is:

$$f(x|\theta) = \frac{1}{(2\pi\theta)^{1/2}} \exp\left(-\frac{x^2}{2\theta}\right)$$

The log density is:

$$\log f(x|\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\theta) - \frac{x^2}{2\theta}$$

the first and second derivatives are:

$$\frac{d}{d\theta} \log f(x|\theta) = -\frac{1}{2\theta} + \frac{x^2}{2\theta^2} = \frac{x^2 - \theta}{2\theta^2}$$

$$\frac{d^2}{d\theta^2} \log f(x|\theta) = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3}$$

Example: Normal log likelihood

$$\ell_n(\theta) = \sum_{i=1}^n \log f(X_i|\theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\theta) - \frac{1}{2\theta} \sum_{i=1}^n X_i^2$$

$$\frac{d}{d\theta} \ell_n(\theta) = -\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n X_i^2$$

The MLE is $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i^2$

Example: Score and Fisher Information

$$\frac{d}{d\theta} \log f(x|\theta) = \frac{x^2 - \theta}{2\theta^2}$$

The efficient score $S = \frac{x^2 - \theta_0}{2\theta_0^2}$

$$\text{var}[S] = \frac{\mathbb{E}[(X^2 - \theta_0)^2]}{4\theta_0^4} = \frac{\mathbb{E}[X^4 - 2X^2\theta_0 + \theta_0^2]}{4\theta_0^4} = \frac{3\theta_0^2 - 2\theta_0^2 + \theta_0^2}{4\theta_0^4} = \frac{1}{2\theta_0^2}$$

Example: Information Equality $\mathcal{I}_\theta = \mathcal{H}_\theta$

$$\mathcal{H}_\theta = \mathbb{E}\left[-\frac{d^2}{d\theta^2} \log f(X|\theta_0)\right] = \mathbb{E}\left[-\frac{d}{d\theta} \frac{x^2 - \theta_0}{2\theta_0^2}\right] = \mathbb{E}\left[\frac{2x^2 - \theta_0}{2\theta_0^3}\right] = \frac{1}{2\theta_0^2}$$

$$\mathcal{I}_\theta = \text{var}[S] = \frac{1}{2\theta_0^2} = \mathcal{H}_\theta$$

The total Fisher information for n observations is $I_n(\theta_0) = n \times \mathcal{I}_\theta = \frac{n}{2\theta_0^2}$

The variance of $\hat{\theta}$ is $\frac{1}{I_n(\theta_0)} = 2\theta_0^2/n$

So we can get the plug-in estimator for the standard error as $\sqrt{2\hat{\theta}^2/n}$.

Simulation: Fisher Information Standard Errors

```
for(i in 1:100000){  
  x <- rnorm(100, sd=sqrt(5))  
  thetahat[i] <- sum(x^2)/100  
  diff[i] <- thetahat[i]-5  
  se[i] <- sqrt(2*thetahat[i]^2/100) }  
mean(thetahat)  
5.002307  
sqrt(var(diff))  
0.7102136  
mean(se)  
0.707433
```

What If the Model Is Wrong? Robust Variance

Under correct specification, $\mathcal{I}_\theta = \mathcal{H}_\theta$ and either one gives the variance.

Under misspecification, they diverge:

- $\mathcal{H}$ (curvature) still captures how the estimator behaves locally.
- $\mathcal{I}$ (score variance) captures the actual variability in the data.
- The correct variance requires *both*:

$$\hat{\mathbf{V}} = \hat{\mathcal{H}}^{-1} \hat{\mathcal{I}} \hat{\mathcal{H}}^{-1}$$

This is the **sandwich formula** — the same structure as the heteroskedasticity-robust variance from Lecture 6, now derived from likelihood theory.

Summary: Scores, Information, and Efficiency

These four concepts are **not specific to MLE** — they recur throughout estimation theory:

- 1 Score** → the derivative of the log-density. Setting it to zero gives estimating equations — the same objects that define moment conditions and influence functions.
- 2 Fisher Information** → the variance of the score; measures how much the data can tell you about θ .
- 3 Efficiency bound** → no unbiased estimator can beat $(n\mathcal{I}_\theta)^{-1}$. MLE achieves this when the likelihood is correctly specified.
- 4 Likelihood** → when a full parametric model is available, the likelihood gives us a concrete score and a concrete bound.

Next: apply this to the normal linear model and show that OLS *is* the MLE.

Likelihood methods for linear model

- The Normal Regression model assumes that $y \sim N(\mu_Y, \sigma^2)$, or that $e \sim N(0, \sigma^2)$.

$$\begin{aligned} f(y|\mathbf{x}) &= \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(y - \mathbf{x}'\beta)^2\right) \\ f(y_1, \dots, y_n | \mathbf{x}_1, \dots, \mathbf{x}_n) &= \prod_{i=1}^n f(y_i | x_i) \\ &= \prod_{i=1}^n \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mathbf{x}_i'\beta)^2\right) \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i'\beta)^2\right) \\ &\equiv \mathcal{L}_n(\beta, \sigma^2) \end{aligned}$$

Log Likelihood of the normal linear model

$$\log \mathcal{L}_n(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2 \equiv \ell_n(\beta, \sigma^2)$$

Classic Normal Regression Model

- The Classic Normal Regression Model consists of the following assumptions:

- 1 $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$

- 2 $\mathbf{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$

- or equivalently: $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$

- 3 $\text{rank}(\mathbf{X}) = K$.

In this model, $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the multivariate normal density function whose pdf is:

$$f(\mathbf{e}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \mathbf{e}'\mathbf{e}\right)$$

Background on Multivariate Normal

- A multivariate normal's parameters are a vector of means and a variance covariance matrix.
- Linear functions of a multinormal vector $\mathbf{y}$ are also normal.
- If $\mathbf{z} = \mathbf{g} + \mathbf{H}\mathbf{y}$, where $\mathbf{g}$ and $\mathbf{H}$ are non-random, and $\mathbf{H}$ has full row rank, then

$$\mathbf{z} \sim N(\mathbf{g} + \mathbf{H}\boldsymbol{\mu}, \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}')$$

- In the case of MVN, independence is the same as uncorrelated.

Roadmap: MLE as a Special Case

- We have the normal linear model: $\mathbf{y}|\mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$.
- Solve for the MLE and show it equals OLS — the full likelihood buys us nothing new for $\hat{\boldsymbol{\beta}}$.
- The **score** of the normal model *is* the OLS moment condition $\mathbf{X}'\mathbf{e} = 0$. This is the key link: MLE is just-identified GMM with score moments.
- The payoff of normality: exact finite-sample distributions (t, χ^2, F) that do not require asymptotics.

Solving for the MLE

- Given the likelihood, we can solve for the MLE $(\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2)$

$$\frac{\partial}{\partial \beta} \left[-\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2 \right]$$

$$\frac{2}{2\hat{\sigma}_{MLE}^2} \sum_{i=1}^n X_i (Y_i - \mathbf{x}_i' \hat{\beta}_{MLE}) = 0$$

$$\frac{\partial}{\partial \sigma^2} \left[-\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2 \right]$$

$$-\frac{n}{2} \frac{1}{\hat{\sigma}_{MLE}^2} + \frac{1}{2\hat{\sigma}_{MLE}^4} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \hat{\beta}_{MLE})^2 = 0$$

Overall Likelihood

- $\hat{\beta}_{MLE} = (\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i')^{-1} (\sum_{i=1}^n \mathbf{x}_i Y_i) = \hat{\beta}_{ols}.$
- $\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \hat{\beta}_{MLE})^2.$ Note: this divides by n , not $n-k$, so the MLE is **biased** for σ^2 . The unbiased estimator s^2 divides by $n-k$.
- $\log \mathcal{L}(\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2) = -\frac{n}{2} \log(2\pi \hat{\sigma}_{MLE}^2) - n/2$
- You will see the "log likelihood" reported as a measure of fit, `logLik()`
- The Akaike's AIC is $-2\log\text{-likelihood} + 2n_{par}$, weighing model performance versus complexity.
- AIC gives an asymptotically unbiased estimator of the expected relative Kullback-Leibler divergence when approximating an unknown distribution and a given model.

R Example: Log-Likelihood and AIC

```
data(swiss)
mod1 <- lm(Fertility ~ Education, data = swiss)
mod2 <- lm(Fertility ~ Education + Agriculture,
           data = swiss)
mod3 <- lm(Fertility ~ Education + Agriculture
           + Catholic + Infant.Mortality, data=swiss)
# Log-likelihoods (higher = better fit)
sapply(list(mod1, mod2, mod3), logLik)
# -168.3 -166.0 -155.3
# AIC = -2*logLik + 2*k (lower = better)
sapply(list(mod1, mod2, mod3), AIC)
# 342.5 340.0 322.6
```

AIC penalizes complexity: mod3 wins because the likelihood improvement outweighs the $2k$ penalty.

Score of the Normal Regression Model (Hansen 5.14)

The likelihood scores are the derivatives of the log-likelihood:

$$\frac{\partial}{\partial \beta} \ell_n(\beta, \sigma^2) = \frac{1}{\sigma^2} \sum_{i=1}^n X_i (Y_i - X_i' \beta) = \frac{1}{\sigma^2} \mathbf{X}' \mathbf{e}$$

$$\frac{\partial}{\partial \sigma^2} \ell_n(\beta, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (Y_i - X_i' \beta)^2$$

Setting the score for β to zero:

$$\frac{1}{\sigma^2} \mathbf{X}' \mathbf{e} = 0 \quad \Longleftrightarrow \quad \mathbf{X}' \mathbf{e} = 0 \quad \Longleftrightarrow \quad \mathbf{X}' (\mathbf{Y} - \mathbf{X} \beta) = 0$$

These are exactly the **OLS normal equations**! The MLE for β equals the OLS estimator because the score FOC is proportional to the least squares FOC.

Scores as Moment Conditions

The score for β evaluated at the true parameter is:

$$S_i(\beta_0) = \frac{\partial}{\partial \beta} \log f(Y_i | X_i, \beta_0, \sigma^2) = \frac{1}{\sigma^2} X_i e_i$$

The population moment condition from the CEF is $\mathbb{E}[X_i e_i] = 0$.

Key observation: Setting the score to zero in the sample,

$$\frac{1}{n} \sum_{i=1}^n S_i(\hat{\beta}) = 0 \quad \Longleftrightarrow \quad \frac{1}{n} \sum_{i=1}^n X_i \hat{e}_i = 0$$

is the **same** as solving the sample moment condition $\frac{1}{n} \sum X_i (Y_i - X_i' \hat{\beta}) = 0$.

MLE, OLS, and method of moments all give the *same* estimator for β in the normal linear model.

Information Matrix of Normal Regression (Hansen 5.14)

The Fisher information matrix for the normal regression model is:

$$\mathcal{I} = \text{var} \begin{bmatrix} \frac{\partial}{\partial \beta} \ell(\beta, \sigma^2) \\ \frac{\partial}{\partial \sigma^2} \ell(\beta, \sigma^2) \end{bmatrix} = \begin{pmatrix} \frac{1}{\sigma^2} \mathbf{X}'\mathbf{X} & \mathbf{0} \\ \mathbf{0} & \frac{n}{2\sigma^4} \end{pmatrix}$$

- The matrix is **block diagonal**: estimation of β and σ^2 are independent.
- The Cramér-Rao lower bound for β is:

$$\mathcal{I}_{\beta}^{-1} = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$$

which is exactly the variance of $\hat{\beta}_{OLS}$.

- OLS **achieves the Cramér-Rao bound** — it is efficient among all unbiased estimators under normality.

From Scores to the Sandwich (connecting to Lecture 6)

Under **correct specification** ($e \sim N(0, \sigma^2)$):

$$\mathcal{I}_\theta = \mathcal{H}_\theta \implies V = \mathcal{H}^{-1} = \mathcal{G}^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

One formula suffices. This is the classical OLS variance.

Under **misspecification** (heteroskedasticity, non-normality):

$$\mathcal{I}_\theta \neq \mathcal{H}_\theta \implies V = \mathcal{H}^{-1} \mathcal{G} \mathcal{H}^{-1}$$

- The “bread” $\mathcal{H}^{-1} \rightarrow (\mathbf{X}'\mathbf{X})^{-1}$
- The “meat” $\mathcal{G} \rightarrow \mathbf{X}'\mathbf{D}\mathbf{X}$ where $D = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$
- The sandwich: $(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{D}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$

This is the **heteroskedasticity-robust variance** from Lecture 6 (HC0–HC3)!

The “sandwich” estimator is the likelihood-based variance under misspecification.

MLE as a Special Case of Moment-Based Estimation

Maximum likelihood solves k score equations in k unknowns: $\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(Y_i|X_i, \theta) = 0$

GMM solves m moment conditions in k unknowns: $\frac{1}{n} \sum_{i=1}^n g(Y_i, X_i, \theta) = 0$

- When $m = k$ and $g = \text{score}$: **GMM = MLE**. MLE is just-identified GMM.
- When $m > k$: GMM uses an *optimal weighting matrix* to combine them.
- IV example: $\mathbb{E}[Z_i e_i] = 0$ gives more moment conditions than parameters.

Key insight: $\text{MLE} \subset \text{GMM}$. Score equations are one choice of moment conditions. When the likelihood is misspecified, the sandwich variance replaces the information-based one.

Summary: The Score-Based View

Concept	Formula	Role
Score	$S_i = \frac{1}{\sigma^2} X_i e_i$	FOC $\rightarrow$ OLS normal eqns
Information	$\mathcal{I}_\beta = \frac{1}{\sigma^2} \mathbf{X}' \mathbf{X}$	Precision of $\hat{\beta}$
CR Bound	$\sigma^2 (\mathbf{X}' \mathbf{X})^{-1}$	OLS achieves it
Sandwich	$(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' D \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1}$	Robust to misspec.
GMM link	Score = moment cond.	MLE $\subset$ GMM

Next: exact finite-sample distributions under normality — $\hat{\beta}$, s^2 , t , and F .

The Payoff of Normality: Exact Distributions

So far we have shown:

- OLS = MLE under normality, achieving the Cramér-Rao bound.
- The sandwich variance connects likelihood theory to robust inference.

Now: the normality assumption buys us something no other approach can — **exact finite-sample distributions** for every quantity of interest:

- 1 $\hat{\beta} \sim N$ (linear function of normal $\mathbf{e}$)
- 2 $s^2 \sim \chi^2$ (quadratic form in normal $\mathbf{e}$)
- 3 $T \sim t_{n-k}$ (ratio of independent normal and χ^2)
- 4 $F \sim F_{q,n-k}$ (likelihood ratio test)

These hold for *any* n — no large-sample approximation needed.

Proof of normality of $\hat{\beta}$ in CNRM

We can show that $\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e})$$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$$

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$$

$$\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$$

$$\text{Rank}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \text{Rank}(\mathbf{X}) = K$$

So $\hat{\beta} - \beta$ is a full-row-rank linear transformation of the multinormal vector $\mathbf{e}$

$$\hat{\beta} - \beta \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})$$

Distribution of Residuals (Hansen 5.7)

Recall $\hat{\mathbf{e}} = \mathbf{M}\mathbf{e}$ where $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Since $\mathbf{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ and $\hat{\mathbf{e}}$ is a linear function of $\mathbf{e}$:

$$\hat{\mathbf{e}}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{M})$$

Key fact: $\hat{\beta}$ and $\hat{\mathbf{e}}$ are **independent** (conditional on $\mathbf{X}$).

Proof: Their joint covariance is

$$\text{Cov}(\hat{\beta} - \beta, \hat{\mathbf{e}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \sigma^2 \mathbf{I} \cdot \mathbf{M} = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{M}}_{=0} = \mathbf{0}$$

Since they are jointly normal and uncorrelated, they are independent.

Why this matters: It means $s^2 = \hat{\mathbf{e}}'\hat{\mathbf{e}}/(n - k)$ is independent of $\hat{\beta}$. This is what allows us to form the t -statistic as a ratio of independent normal and χ^2 components.

Distribution of residual sum of squares $s_{\hat{\mathbf{e}}}^2 = \frac{\hat{\mathbf{e}}' \hat{\mathbf{e}}}{(N-K)}$ in CNRM

Thm: Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$, let $\mathbf{A}$ be idempotent. Then $\mathbf{z}' \mathbf{A} \mathbf{z} \sim \chi^2(\nu)$ where $\nu = \text{Rank}(\mathbf{A})$.

$$\begin{aligned}\mathbf{e} | \mathbf{X} &\sim N(\mathbf{0}, \sigma^2 \mathbf{I}) \\ \mathbf{e} / \sigma | \mathbf{X} &\sim N(\mathbf{0}, \mathbf{I})\end{aligned}$$

Recall $\hat{\mathbf{e}} = \mathbf{M} \mathbf{e}$, so:

$$\begin{aligned}(N-K)s_{\hat{\mathbf{e}}}^2 &= \hat{\mathbf{e}}' \hat{\mathbf{e}} = \mathbf{e}' \mathbf{M} \mathbf{e} \\ \frac{\mathbf{e}'}{\sigma} \mathbf{M} \frac{\mathbf{e}}{\sigma} &\sim \chi^2(N-K) \quad (\mathbf{M} \text{ is idempotent, rank } N-K)\end{aligned}$$

That is, in a linear regression model, $\frac{(N-K)s_{\hat{\mathbf{e}}}^2}{\sigma^2} \sim \chi_{n-k}^2$

R Example: χ^2 Distribution of s^2

```
set.seed(1); n <- 50; k <- 3; sigma2 <- 4
X <- cbind(1, matrix(rnorm(n*(k-1)), n, k-1))
beta <- c(2, -1, 0.5)
scaled_s2 <- replicate(10000, {
  y <- X %*% beta + rnorm(n, sd = sqrt(sigma2))
  ehat <- resid(lm(y ~ X - 1))
  sum(ehat^2) / sigma2      # (n-k)*s^2 / sigma^2
})
```

```
# Compare to chi-squared(n-k)
mean(scaled_s2)      # ~47 (= n-k)
var(scaled_s2)       # ~94 (= 2*(n-k))
```

Theory: $\mathbb{E}[\chi_{47}^2] = 47$,
 $\text{Var}[\chi_{47}^2] = 94$.

The simulation matches.

From Known σ to Estimated s : Why Heavier Tails

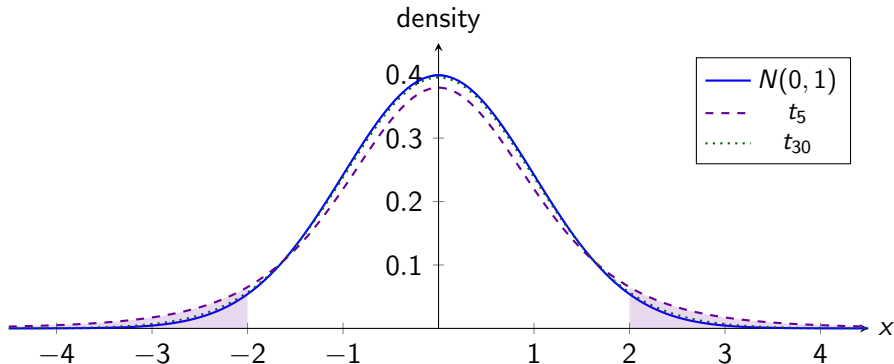
If σ were known, the j -th coefficient is exactly normal:

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0, 1)$$

But σ is unknown. Replacing it with s introduces extra randomness:

- When s underestimates σ , the ratio is *inflated* — producing occasional large values.
- When s overestimates, the ratio is compressed.
- Net effect: **heavier tails** than the normal, especially when $n-k$ is small (because s is noisy with few degrees of freedom).
- As $n-k \rightarrow \infty$, $s \rightarrow \sigma$ and $t_{n-k} \rightarrow N(0, 1)$.

t -Distribution vs. Normal: Heavier Tails



With few degrees of freedom (t_5), the tails carry substantially more mass beyond ± 2 than the normal.

From Normal to t : Estimated Variance

In practice, we do not know the true error variance σ^2 . The unbiased estimator is:

$$s_{\hat{e}}^2 = \frac{1}{n-k} \sum_{i=1}^n \hat{e}_i^2 = \frac{\hat{\mathbf{e}}' \hat{\mathbf{e}}}{n-k}$$

We plug in this estimate to standardize:

$$T = \frac{\hat{\beta}_j - \beta_j}{\sqrt{s_{\hat{e}}^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}}$$

This is no longer standard normal, but it has a t -**distribution**.

The t Statistic and its Distribution

Recall: if $Z \sim N(0, 1)$, $W \sim \chi^2(n - k)$, and $Z \perp W$, then $\frac{Z}{\sqrt{W/(n-k)}} \sim t(n - k)$.

In our case, $Z = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0, 1)$ and $W = \frac{(n-k)s_{\hat{e}}^2}{\sigma^2} \sim \chi^2(n - k)$.

Taking the ratio and simplifying (σ^2 cancels):

$$T = \frac{\hat{\beta}_j - \beta_j}{\sqrt{s_{\hat{e}}^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim t(n - k)$$

Simulation: t -Statistics Match the t -Distribution

```
set.seed(1); n <- 15; k <- 3; sigma2 <- 4
X <- cbind(1, matrix(rnorm(n*(k-1)), n, k-1))
beta <- c(2, -1, 0.5)
t_stats <- replicate(10000, {
  y <- X %*% beta + rnorm(n, sd = sqrt(sigma2))
  summary(lm(y ~ X - 1))$coefficients[2, "t value"]
})
```

```
mean(abs(t_stats) > 2)    # 0.067
2*pt(-2, df = n-k)       # 0.065
2*pnorm(-2)               # 0.046
```

With $n-k = 12$, using $N(0, 1)$ critical values rejects 4.6% instead of 6.5%.

Limits of the t -Statistic

- The t -statistic $T = \frac{\hat{\beta} - \beta_0}{s(\hat{\beta})}$ follows t_{n-k} **exactly** under two conditions: homoskedasticity and i.i.d. normality of e .
- Without normality, OLS is still unbiased, but the exact t_{n-k} distribution no longer holds.
- With HC0–HC3 robust standard errors, we do not have an exact finite-sample distribution for T .
- **What do we do instead?** We rely on *large-sample approximations*: the CLT guarantees that $T \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$, regardless of the error distribution. This is the subject of Lectures 10–11 (Asymptotics).
- For now: the normal model gives exact, clean results. Relaxing normality requires asymptotic tools.

The F -test as a Likelihood Ratio Test (Hansen 5.13)

Consider testing $\mathbb{H}_0 : \beta_2 = 0$ in $Y = X_1'\beta_1 + X_2'\beta_2 + e$.

The likelihood ratio statistic compares the maximized log-likelihoods:

$$\text{LR} = 2 \left(\ell_n(\hat{\beta}, \hat{\sigma}^2) - \ell_n(\tilde{\beta}_1, \tilde{\sigma}^2) \right) = n \log \left(\frac{\tilde{\sigma}^2}{\hat{\sigma}^2} \right)$$

where $\tilde{\sigma}^2$ is from the restricted (null) model. This is equivalent to the F -statistic:

$$F = \frac{(\tilde{\sigma}^2 - \hat{\sigma}^2)/q}{\hat{\sigma}^2/(n-k)} \sim F_{q, n-k} \quad \text{under } \mathbb{H}_0$$

- $q = \dim(\beta_2)$ is the number of restrictions.
- Under the null, F has an *exact* F -distribution in the normal model.
- This justifies the F -test reported by `anova()` and regression output.
- The t -test is a special case: when $q = 1$, $F = T^2$ and $F_{1, n-k} = t_{n-k}^2$.

Summary: Small-Sample Distributions

Under the classical normal regression model:

- $\hat{\beta}|\mathbf{X} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ *(linear fn of normal)*
- $\hat{\mathbf{e}}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{M}), \quad \hat{\beta} \perp \hat{\mathbf{e}}$ *(projection)*
- $\frac{(n-k)s^2}{\sigma^2} \sim \chi_{n-k}^2$ *(idempotent quadratic form)*
- $T = \frac{\hat{\beta}_j - \beta_j}{s(\hat{\beta}_j)} \sim t_{n-k}$ *(normal / $\sqrt{\chi^2/df}$)*
- $F = \frac{(\tilde{\sigma}^2 - \hat{\sigma}^2)/q}{\hat{\sigma}^2/(n-k)} \sim F_{q,n-k}$ *(likelihood ratio)*

These **exact** results hold in finite samples. Without normality, we rely on asymptotic approximations.

Confidence Intervals

- $\hat{\beta}$ is a **point estimate** for a coefficient β .
- We can instead estimate an interval, $\hat{C} = [\hat{L}, \hat{U}]$ which contains the true value with high probability.
- An interval estimate $\hat{C}$ is called a $1 - \alpha$ confidence interval when $Pr(\beta \in \hat{C}) = 1 - \alpha$. The value $1 - \alpha$ is called the **coverage probability**.
- The key mistake is in thinking that the above statement treats β as random and $\hat{C}$ is fixed, (the probability that β is in $\hat{C}$).
- $Pr(\beta \in \hat{C})$ is the probability that the random set $\hat{C}$ covers or contains β .

$$\hat{C} = [\hat{\beta} - c \cdot s(\hat{\beta}), \hat{\beta} + c \cdot s(\hat{\beta})]$$

Confidence Intervals in Practice

$$Pr(\beta \in \hat{C}) = Pr(-c \leq T(\beta) \leq c)$$

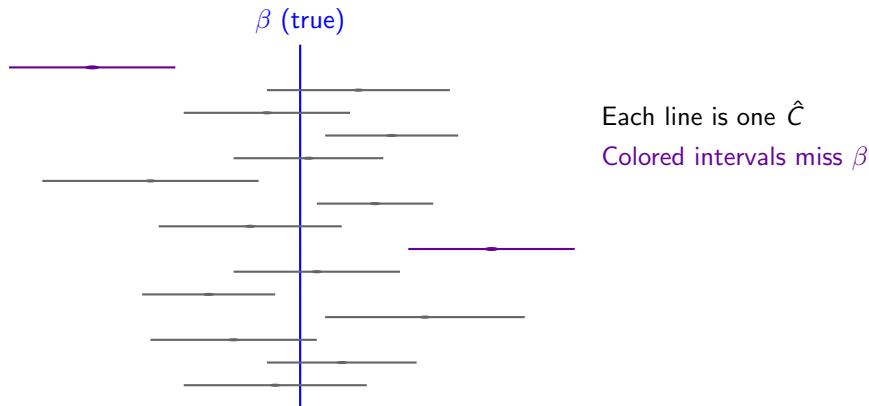
$$Pr(\beta \in \hat{C}) = 2F(c) - 1$$

Our goal is to set this coverage probability equal to $1 - \alpha$, or $F(c) = 1 - \alpha/2$.

If $\alpha = .05$, we solve $c = F^{-1}(1 - .05/2)$. In case of a normal, $c = 1.96 \approx 2$

$$\hat{C} = [\hat{\beta} - 2 \cdot s(\hat{\beta}), \hat{\beta} + 2 \cdot s(\hat{\beta})]$$

Confidence Intervals: The Random Interval Covers a Fixed Parameter



β is *fixed*. Each sample yields a different random interval $\hat{C}$. About $1-\alpha$ of them cover β .

t test and p-values

- A theory is said to have *testable implications* if it can be falsified.
- For example, a theory may be false if $\beta = \beta_0$. This is called a "null hypothesis" $\mathbb{H}_0$.
- We further specify the complement of $\mathbb{H}_0$ as $\mathbb{H}_1$.
- A statistic can be informative, some realizations may be unlikely if $\mathbb{H}_0$ is true.
- Define a test statistic: $|T| = \left| \frac{\hat{\beta} - \beta_0}{s(\hat{\beta})} \right|$ and set a critical value c .

Reject $\mathbb{H}_0$ if $|T| > c$

- A p-value indexes a test's strength of rejection of the null.
- In a normal regression model, $p = 2(1 - F_{n-k}(|T|))$

Example: Swiss Fertility Data

	<i>Dep. var: Fertility</i>
Education (%)	−0.98*** (0.15)
Agriculture (%)	−0.15** (0.07)
Catholic (%)	0.12*** (0.03)
Infant.Mortality (%)	1.08*** (0.38)
Constant	62.10*** (9.60)
Observations	47
R^2	0.70
Residual Std. Error	7.17 (df = 42)
Note:	* p<0.1; ** p<0.05; *** p<0.01

Did education share predict the average fertility across Swiss districts in 1888?

Focus on the Education coefficient: $\hat{\beta} = -0.98$, $se(\hat{\beta}) = 0.15$.

The t -statistic is $-0.98/0.15 = -6.61$, with 42 degrees of freedom.

Is this large enough to reject $\mathbb{H}_0 : \beta = 0$?

Example: Hypothesis Test

$$\hat{\beta} = -0.98, \quad \beta_0 = 0, \quad se(\hat{\beta}) = 0.15, \quad Df = 47 - 5 = 42$$

$$t = \frac{\hat{\beta} - \beta_0}{se(\hat{\beta})} = \frac{-0.98 - 0}{0.15} = -6.61$$

$$c = qt(1 - .025, 42) = 2.02$$

$$|t| = 6.61 > 2.02 = c \implies \text{Reject } \mathbb{H}_0$$

$$p = 2(1 - pt(6.61, 42)) = 5.2 \times 10^{-8}$$

For each 1 percentage point increase in post-primary attendance, fertility is estimated to be 0.98 pp lower, with a 95% CI of approximately (0.7, 1.3).

Beyond Single Coefficients: Joint Restrictions

- The t -test asks: is *one* coefficient zero?
- Many research questions involve *multiple* coefficients simultaneously.
- **Example:** Does a job training program affect wages?

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Training}_i + \beta_2 (\text{Training}_i \times \text{Experience}_i) + \beta_3 \text{Experience}_i + e_i$$

- The effect of training is $\beta_1 + \beta_2 \cdot \text{Experience}_i$ — it depends on *both* β_1 and β_2 .
- To test whether training has *any* effect, we need:

$$\mathbb{H}_0 : \beta_1 = 0 \text{ and } \beta_2 = 0$$

- A t -test on β_1 alone misses the interaction; a t -test on β_2 alone misses the main effect. We need a **joint test**.

The F -Test for Joint Restrictions

To test $\mathbb{H}_0 : \mathbf{R}\beta = \mathbf{r}$ where $\mathbf{R}$ is $q \times k$:

- 1 Estimate the **unrestricted** model: get $\hat{\mathbf{e}}'\hat{\mathbf{e}}$ (the full SSR).
- 2 Estimate the **restricted** model (imposing $\mathbb{H}_0$): get $\tilde{\mathbf{e}}'\tilde{\mathbf{e}}$.
- 3 The F -statistic measures how much fit deteriorates under the restriction:

$$F = \frac{(\tilde{\mathbf{e}}'\tilde{\mathbf{e}} - \hat{\mathbf{e}}'\hat{\mathbf{e}})/q}{\hat{\mathbf{e}}'\hat{\mathbf{e}}/(n-k)} \sim F_{q, n-k} \quad \text{under } \mathbb{H}_0$$

Intuition: If $\mathbb{H}_0$ is true, dropping the restricted variables should not increase SSR much. A large F means the restricted model fits **substantially worse** $\rightarrow$ reject.

When $q = 1$, this reduces to $F = T^2$ — the t -test is a special case.

Example: Joint Test for Treatment Effect

```
set.seed(42); n <- 80
experience <- runif(n, 1, 20)
training <- rbinom(n, 1, 0.5)
wage <- 10 + 3*training + 0.5*training*experience +
        0.8*experience + rnorm(n, sd = 3)

full <- lm(wage ~ training*experience)           # unrestricted
restricted <- lm(wage ~ experience)              # restricted

anova(restricted, full)  # joint F-test
```

Example: Why the Joint Test Matters

```
summary(full)$coefficients[c("training",  
                             "training:experience"), ]  
  
#           Estimate Std.Err t      Pr(>|t|)  
# training           1.81    1.48  1.22    0.225  
# training:experience  0.60    0.13  4.64   <0.001  
  
anova(restricted, full)  
# F = 39.2, df1 = 2, df2 = 76, p < 0.001
```

- The t -test on training alone: $p = 0.225$ — not significant!
- But the joint F -test: $p < 0.001$ — training clearly matters.
- The interaction absorbs much of the treatment's impact at higher experience levels, so the main effect alone is small.
- **Lesson:** When a variable enters with an interaction, always test jointly.

Writing Joint Restrictions as $R\beta = r$

For the model $\text{Wage} = \beta_0 + \beta_1 \text{Tr} + \beta_2 \text{Tr} \times \text{Exp} + \beta_3 \text{Exp} + e$:

$$\mathbb{H}_0 : \beta_1 = 0 \text{ and } \beta_2 = 0 \quad \Longleftrightarrow \quad \underbrace{\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_R \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ 0 \end{pmatrix}}_r$$

The F -statistic can also be written directly from the unrestricted estimates:

$$F = \frac{(R\hat{\beta} - r)'[R\hat{V}R']^{-1}(R\hat{\beta} - r)}{q}$$

where $\hat{V} = s^2(\mathbf{X}'\mathbf{X})^{-1}$. This form requires only the *unrestricted* model — no need to re-estimate under $\mathbb{H}_0$.

General: any set of q linear restrictions on β can be tested this way.